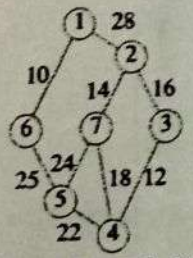
NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END-SEMESTER EXAMINATION - MAY, 2024
B. Tech (Computer Science & Engineering) IVth Semester **(SECTION – A, B, and C)**
Course: Design and Analysis of Algorithms Max. Marks:60 Max. Time: 3 Hrs.
Course Code: CS44115 (for B.Tech-CSE and DD-CSE-CS & DS) / CS47101 (for DD-MA-MCT)

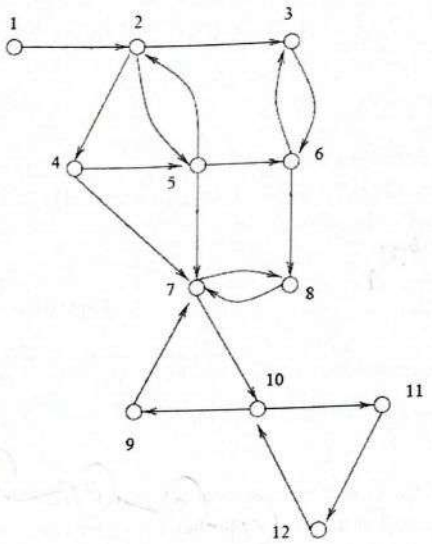
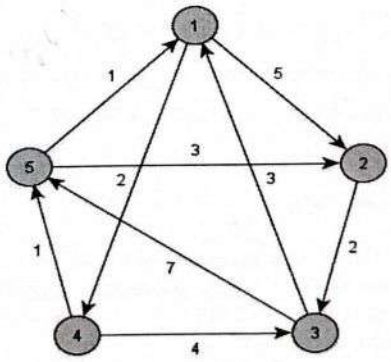
Instruction:

1. Answer all questions.
2. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

Q. No	Question	Marks	CO	BL

1	a. What is the complexity of the below program:
	`void function(int n)`
	`{`
	`int i, j, k , count =0;`
	`for(i=n/2; i<=n; i++)`
	`for(j=1; j + n/2<=n; j= j++)`
	`for(k=1; k<=n; k= k * 2)`
	`count++;`
	`}`
		5M	CO1	Analyze

	b. Suppose we're using quick sort to process data that we're receiving from a connection in a networked system. We want to cover our system from the possibility of being "sabotaged" by hostile connections—we could receive data that is specifically crafted to cause quick sort to have its worstcase performance and thus make our system consume excessive resources and time. Assuming that quick sort simply chooses the first element as the pivot, what is the arrangement of data that produces the worst-case performance in quick sort?	5M	CO1 & CO2	Understand Apply
2	a. For the graph given below, find a Minimum Spanning Tree through Prim's algorithm by taking '1' as the starting node. Show all the stages in the diagrammatic form only.  	6M	CO4	Understand Apply Evaluate
	b. Show the timing analysis of Kruskal's & Prim's Algorithms.	4M	CO1	Analyze
3	a. Solve the below 0/1 knapsack problem using the Least Cost Branch and Bound method. The knapsack capacity is 15. Show the complete state space tree to solve this problem and mention which items are loaded to maximize the profit. **Profit**: 10, 10, 12, 18 **Weight**: 2, 4, 6, 9	5M	CO5	Understand Apply Evaluate
	b. What is the main purpose of studying NP-Completeness? What is the difference between NP Hard and NP-Complete problem? Proof that Clique Decision Problem (CDP) is a NP Hard Problem.	5M	CO5	Understand Create

2.	Define strongly connected component in directed graph? Consider the following graph and compute the DFS tree of the graph. Identify back edges [take vertex 1 as source]. Find all the strongly connected components in the graph. Show all the necessary steps.  	2+4+ 2+4	Understand, Apply, Evaluate
3.	Define P, NP, NP hard, and NP complete problem? Draw the corresponding Ven- diagram to show the relationship among them. The problem of determining whether there exists a cycle in an undirected graph is in P or NP or in both. Justify your answer.	6+2 +2+2	CO-2-3 & Understand
4.	State n-queen problem. Write the Backtracking algorithm to solve 4-queens problem. Apply the algorithm to solve 4-queen problem and draw the corresponding state space tree.	2+6+ 4	
5.	Find single source shortest distance from source vertex 1 to all other vertices using Dijkstra's Algorithm. Show all the intermediate steps. What is the time complexity of the algorithm if you use heap data structure? If you don't use heap data structure then what is the time complexity? Which one is better and when in terms of time complexity.  	8+2+ 2+ 2	CO-3 & Understand Apply

********** ALL THE BEST **********

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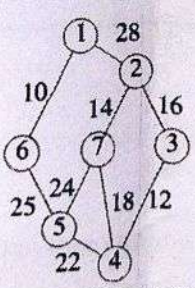
Course Code: CS44115 (for B.Tech-CSE and DD-CSE-CS & DS) / CS47101 (for DD-MA-MCT)

Instruction:

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1	a. What is the complexity of the below program:
	`void function(int n)`
	`{`
	`  int i, j, k , count =0;`
	`  for(i=n/2; i<=n; i++)`
	`  for(j=1; j + n/2<=n; j= j++)`
	`  for(k=1; k<=n; k= k * 2)`
	`  count++;`
	`}`
		5M	CO1	Analyze

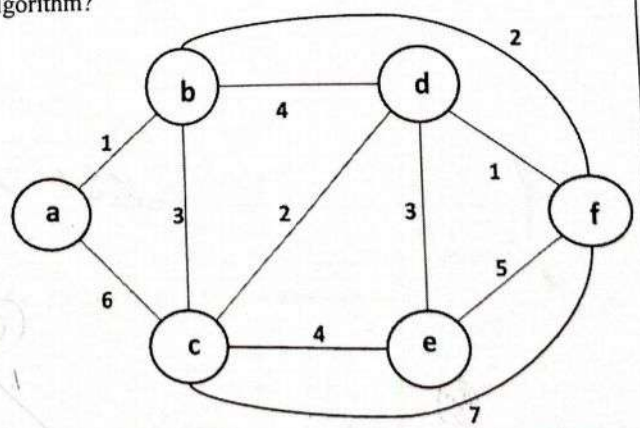
	b. Suppose we're using quick sort to process data that we're receiving from a connection in a networked system. We want to cover our system from the possibility of being "sabotaged" by hostile connections—we could receive data that is specifically crafted to cause quick sort to have its worstcase performance and thus make our system consume excessive resources and time. Assuming that quick sort simply chooses the first element as the pivot, what is the arrangement of data that produces the worst-case performance in quick sort?	5M	CO1 & CO2	Understand Apply
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NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END-SEMESTER EXAMINATION – Jan-June, 2025
CS44115/CS47101 – Design and Analysis of Algorithms
(B.Tech-CSE and DD-CSE-CS & DS / DD-MA-MCT)

IV Semester

Marks 60

Instruction:
1. Answer all questions.
2. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side of margin.

Q. No	Question	Marks	CO & BL
1.	**a)** What is time complexity of Bellman Ford Algorithm?	2	CO-1-4 Remember
	b) In the Branch and Bound method for solving the 0/1 Knapsack Problem, which of the following statements is TRUE? i) Each node in the search tree represents a partial solution including or excluding certain items. ii) The upper bound at a node is calculated by assuming fractional items can be included. iii) A node is pruned if its upper bound is less than the current best (maximum) profit found so far. iv) All of the above.	2	Understand
	c) For a Given connected undirected graph G = (V, E). Which of the following properties are valid for all spanning trees. P1). Contains exactly |V| - 1 edges. P2). The removal of any edge disconnects it. P3). The addition of any edge creates a cycle. i) Only P2 property is valid. ii) All P1, P2, P3 properties are valid. iii) Only P1 and P3 property is valid. iv) Only P2 and P3 property is valid.	2	Evaluate,
	d) A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency: a: 5 b: 9 c: 12 d: 13 e: 16 f: 45 Note : Each character in input message takes 1 byte. If the compression technique used is Huffman Coding, how many bits will be saved in the message? (i) 224 (ii) 800 (iii) 576 (iv) 324	2	Apply,
	e) Which of the following cannot be the sequence of edges added in that order to find a minimum spanning tree of the graph using Kruskal's algorithm? A) (a-b), (d-f), (b-f), (d-c), (d-e) B) (a-b), (d-f), (d-c), (b-f), (d-e) C) (d-f), (a-b), (d-c), (b-f), (d-e) D) (d-f), (a-b), (b-f), (d-e), (d-c) 	2	Evaluate